CS229 Machine Learning

Computational Problems
Preparation for the Final Exam

Instructions:

- This is a preparation for the **open-book** final exam.
- Preparation consists of short computational problems covering Chapters 1–7 of the PRML textbook.
- Each problem should be solved in max 30 minutes with minimal aids like a basic calculator.
- Provide your answers in a clear, step-by-step manner to demonstrate your reasoning.

Spring 2026

Computing Basic Statistics (Mean, Covariance)

Statement. We have three data points in 2-dimensional space:

$$\mathbf{x}_1 = (1, 2), \quad \mathbf{x}_2 = (2, 3), \quad \mathbf{x}_3 = (3, 5).$$

(a) Compute the sample mean vector of these points.

(b) Compute the 2×2 sample covariance matrix.

(c) Verify that the covariance matrix is positive semidefinite by performing an eigenvalue analysis.

Grading:

- *Sample mean – 3 pts*
- *Covariance matrix – 4 pts*
- *PSD check (eigenvalue analysis) – 3 pts*

Total: 10 points

Solution

(a) **Sample Mean**:

$$\bar{\mathbf{x}} = \frac{1}{3}\big[(1, 2) + (2, 3) + (3, 5)\big] = \left(2,\ \frac{10}{3}\right).$$

(b) **Covariance**:

$$\mathbf{S} = \frac{1}{3}\sum_{n=1}^{3}(\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^\top = \begin{pmatrix} 2/3 & 1 \\ 1 & 14/9 \end{pmatrix}.$$

(c) **Positive semidefinite matrix via Eigenvalue Analysis**: To verify if a symmetric matrix is positive semidefinite, we must show that all of its eigenvalues λ are non-negative ($\lambda \geq 0$). We find the eigenvalues by solving the characteristic equation $\det(\mathbf{S} - \lambda\mathbf{I}) = 0$:

$$\det\begin{pmatrix} \frac{2}{3} - \lambda & 1 \\ 1 & \frac{14}{9} - \lambda \end{pmatrix} = 0$$

Expanding the determinant:

$$\left(\frac{2}{3} - \lambda\right)\left(\frac{14}{9} - \lambda\right) - (1)(1) = 0$$

$$\frac{28}{27} - \frac{2}{3}\lambda - \frac{14}{9}\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - \frac{20}{9}\lambda + \frac{1}{27} = 0$$

Using the quadratic formula to solve for λ:

$$\lambda_{1,2} = \frac{\frac{20}{9} \pm \sqrt{\left(-\frac{20}{9}\right)^2 - 4(1)\left(\frac{1}{27}\right)}}{2} = \frac{\frac{20}{9} \pm \sqrt{\frac{400}{81} - \frac{12}{81}}}{2} = \frac{\frac{20}{9} \pm \frac{\sqrt{388}}{9}}{2} = \frac{10 \pm \sqrt{97}}{9}$$

Since $9^2 < 97 < 10^2$, we know that $\sqrt{97} \approx 9.85$. Therefore, both eigenvalues are positive:

- $\lambda_1 = \frac{10+9.85}{9} \approx 2.205 > 0$
- $\lambda_2 = \frac{10-9.85}{9} \approx 0.017 > 0$

Because all eigenvalues are strictly positive ($\lambda > 0$), the covariance matrix is positive definite, which means it satisfies the condition of being positive semidefinite ($\lambda \geq 0$).

Maximum Likelihood for a Univariate Gaussian

Statement. We have a one-dimensional dataset:

$$\{2.0,\ 2.5,\ 3.5,\ 4.0\}.$$

We assume it comes from a Gaussian $\mathcal{N}(\mu, \sigma^2)$. We want to:

(a) Write down the log-likelihood function.

(b) Compute the maximum-likelihood estimates (MLE) $\hat{\mu}$ and $\hat{\sigma}^2$.

Grading:

- *Log-likelihood expression – 5 pts*
- *MLE for μ, σ^2 – 5 pts*

Total: 10 points

Solution

(a) **Log-likelihood**: For data $\{x_n\}$ in 1D,

$$\ln \mathcal{N}(\mu, \sigma^2) = -\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2,$$

here $N = 4$.

(b) **MLE**:

$$\frac{\partial}{\partial \mu}\Big(-\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\Big) = \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n - \mu) = 0$$

$$\sum_{n=1}^{N} x_n = N\mu \implies \mu = \frac{1}{N}\sum_{n=1}^{N} x_n = \frac{1}{4}(2 + 2.5 + 3.5 + 4) = 3.0$$

$$\frac{\partial}{\partial \sigma^2}\Big(-\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\Big) = -\frac{N}{2}\frac{4\pi}{2\pi\sigma^2} + \frac{2}{2\sigma^4}\sum_{n=1}^{N}(x_n-\mu)^2 =$$

$$-\frac{N}{\sigma^2} + \frac{1}{\sigma^4}\sum_{n=1}^{N}(x_n-\mu)^2 = -N + \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2 = 0$$

$$\sum_{n=1}^{N}(x_n-\mu)^2 = N\sigma^2 \implies \sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n-\mu)^2$$

$$\sigma^2 = \frac{1}{4}\sum_{n=1}^{4}(x_n - 3)^2 = 0.625.$$

Evaluating a Simple Bayesian Update (Bernoulli-Beta)

Statement. We have a Bernoulli variable with parameter θ. Its prior is $\mathrm{Beta}(1, 1)$. We observe 2 successes and 1 failure. We want to:

(a) Write the posterior in Beta form.

(b) Give the new parameters α, β.

(c) Compute the posterior mean.

Grading:

- *Posterior formula – 3 pts*
- *Posterior parameters – 3 pts*
- *Posterior mean – 4 pts*

Total: 10 points

Solution

(a) $\mathrm{Beta}(\alpha_0, \beta_0)$ prior with $\alpha_0 = 1, \beta_0 = 1$. After 2 successes, 1 failure, posterior is $\mathrm{Beta}(1 + 2,\ 1 + 1)$.

(b) So it becomes $\mathrm{Beta}(3, 2)$.

(c) The mean is $\frac{3}{3+2} = 0.6$.

Constructing a Polynomial Design Matrix

Statement. We have three 1D input–output pairs:

$$(x_1, t_1) = (0, 1), \quad (x_2, t_2) = (1, 2), \quad (x_3, t_3) = (2, 5).$$

We fit a polynomial of degree 2:

$$y(x) = w_0 + w_1 x + w_2 x^2.$$

(a) Construct the design matrix $\mathbf{\Phi}$.

(b) Write the normal equations for $\mathbf{w}$.

(c) Solve for w_0, w_1, w_2.

Grading:

- *Design matrix – 5 pts*
- *Normal equations – 3 pts*
- *Solve for parameters – 7 pts*

Total: 15 points

Solution

(a) $\mathbf{\Phi} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}$.

(b) Normal equation: $(\mathbf{\Phi}^\top \mathbf{\Phi})\mathbf{w} = \mathbf{\Phi}^\top \mathbf{t}$.

(c) Solving gives

$$\begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{pmatrix} \mathbf{w} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \\ 22 \end{pmatrix} \implies w_0 = 1,\ w_1 = 0,\ w_2 = 1$$

So $y(x) = 1 + x^2$.

Regularized Polynomial Regression with a Small Dataset

Statement. We have 5 data points in 1D:

$$(x_1, t_1) = (0,\ 0.2),\ (x_2, t_2) = (1,\ 1.1),\ (x_3, t_3) = (2,\ 1.4),\ (x_4, t_4) = (3,\ 2.6),\ (x_5, t_5) = (4,\ 4.5).$$

We fit a polynomial of degree 2,

$$y(x) = w_0 + w_1 x + w_2 x^2,$$

with regularization parameter λ. That is, we minimize

$$\sum_{n=1}^{5} \big[t_n - y(x_n)\big]^2 \ + \ \lambda \sum_{j=0}^{2} w_j^2.$$

(a) Construct the 5×3 design matrix $\mathbf{\Phi}$.

(b) Explain how to form the ridge solution.

(c) Suppose $\lambda = 0.5$. Estimate $\mathbf{w}$ and discuss whether it over-regularizes or not.

Grading:

- *Design matrix – 5 pts*
- *Ridge matrix formation – 5 pts*
- *Approx* $\mathbf{w}$ *and discussion – 5 pts*

Total: 15 points

Solution

(a) **Design matrix $\mathbf{\Phi}$**

$$\mathbf{\Phi} = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 0.2 \\ 1.1 \\ 1.4 \\ 2.6 \\ 4.5 \end{pmatrix}.$$

(b) **Ridge solution**

$$\mathbf{w} = (\mathbf{\Phi}^\top \mathbf{\Phi} + \lambda\, \mathbf{I})^{-1}\, (\mathbf{\Phi}^\top \mathbf{t}).$$

For $\lambda = 0.5$:

$$\mathbf{\Phi}^\top \mathbf{\Phi} + 0.5\, \mathbf{I} = \begin{pmatrix} 5.5 & 10 & 30 \\ 10 & 30.5 & 100 \\ 30 & 100 & 354.5 \end{pmatrix}, \quad \mathbf{\Phi}^\top \mathbf{t} = \begin{pmatrix} 0.2 + 1.1 + 1.4 + 2.6 + 4.5 \\ 0.2 \cdot 0 + 1.1 \cdot 1 + 1.4 \cdot 2 + 2.6 \cdot 3 + 4.5 \cdot 4 \\ 0.2 \cdot 0 + 1.1 \cdot 1 + 1.4 \cdot 4 + 2.6 \cdot 9 + 4.5 \cdot 16 \end{pmatrix} = \begin{pmatrix} 9.8 \\ 29.7 \\ 102.1 \end{pmatrix}$$

Hence

$$\mathbf{w} = \begin{pmatrix} 5.5 & 10 & 30 \\ 10 & 30.5 & 100 \\ 30 & 100 & 354.5 \end{pmatrix}^{-1} \begin{pmatrix} 9.8 \\ 29.7 \\ 102.1 \end{pmatrix}.$$

(c) **(c) Numerical estimate and discussion**

$$\left(\mathbf{\Phi}^\top\mathbf{\Phi} + 0.5\,\mathbf{I}\right)^{-1} \approx \begin{pmatrix} 0.5182 & -0.3477 & 0.0542 \\ -0.3477 & 0.6698 & -0.1595 \\ 0.0542 & -0.1595 & 0.0432 \end{pmatrix}.$$

Then

$$\mathbf{w} \approx \begin{pmatrix} 0.5182 & -0.3477 & 0.0542 \\ -0.3477 & 0.6698 & -0.1595 \\ 0.0542 & -0.1595 & 0.0432 \end{pmatrix} \begin{pmatrix} 9.8 \\ 29.7 \\ 102.1 \end{pmatrix} \approx \begin{pmatrix} 0.285 \\ 0.201 \\ 0.205 \end{pmatrix}.$$

Thus the estimated weights are $(w_0, w_1, w_2) \approx (0.285,\ 0.201,\ 0.205)$. Given $\lambda = 0.5$ is not extremely large compared to the diagonal entries of $\mathbf{\Phi}^\top\mathbf{\Phi}$, it only slightly shrinks the polynomial coefficients. If λ were far larger, the solution would move closer to a small-magnitude $\mathbf{w}$ (underfitting); if λ were near zero, it would chase a best-fit polynomial with no regularization.

Bias-Variance Decomposition at a Single Test Point

Statement. We are evaluating a regression model at a specific test input $\mathbf{x}_*$. The true underlying function yields $h(\mathbf{x}_*) = 2.0$, and the intrinsic target noise variance is $\epsilon^2 = 0.1$.

We train our model on three different random datasets $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$. The corresponding predictions made by these trained models at the test point are:

$$y^{(1)}(\mathbf{x}_*) = 1.5, \quad y^{(2)}(\mathbf{x}_*) = 2.0, \quad y^{(3)}(\mathbf{x}_*) = 2.5$$

(a) Calculate the expected prediction $\bar{y}(\mathbf{x}_*)$ across the datasets.

(b) Calculate the squared bias at this test point.

(c) Calculate the variance of the predictions at this test point.

(d) Calculate the total expected squared loss $\mathbb{E}[L]$ at $\mathbf{x}_*$.

Grading:

- *Expected prediction – 2 pts*
- *Squared bias – 3 pts*
- *Variance – 3 pts*
- *Total expected loss – 2 pts*

Total: 10 points

Solution

(a) **Expected Prediction:** The expected prediction is the simple average of the ensemble predictions:

$$\bar{y} = \frac{1}{3}(1.5 + 2.0 + 2.5) = 2.0$$

(b) **Squared Bias:** The bias is the difference between the expected prediction and the true deterministic function value $h(\mathbf{x}_*)$:

$$(\text{bias})^2 = (\bar{y} - h(\mathbf{x}_*))^2 = (2.0 - 2.0)^2 = 0.0$$

(c) **Variance:** The variance measures how much the individual predictions fluctuate around their own expected value:

$$\text{variance} = \frac{1}{3}\sum_{i=1}^{3}(y^{(i)} - \bar{y})^2 = \frac{1}{3}\left[(1.5 - 2.0)^2 + (2.0 - 2.0)^2 + (2.5 - 2.0)^2\right]$$

$$\text{variance} = \frac{1}{3}[0.25 + 0.0 + 0.25] = \frac{0.5}{3} \approx 0.1667$$

(d) **Total Expected Squared Loss:** The central result of the decomposition states that $\mathbb{E}[L] = (\text{bias})^2 + \text{variance} + \text{noise}$:

$$\mathbb{E}[L] = 0.0 + 0.1667 + 0.1 = 0.2667$$

Bayesian Linear Regression Predictive Distribution

Statement. Suppose we have 1D data $\{(0,1),(1,2),(2,5)\}$. We place a Gaussian prior on $\mathbf{w} = (w_0, w_1, w_2)$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I})$ where $\alpha = 2$. The observation noise variance is $\beta^{-1} = 0.1$. We fit the polynomial $y(x) = w_0 + w_1 x + w_2 x^2$.

(a) Posterior mean $\mathbf{m}_N$.

(b) Posterior covariance $\mathbf{S}_N$.

(c) Predictive mean and variance at $x_* = 1.5$.

Grading:

- *Posterior mean – 5 pts*
- *Posterior covariance – 5 pts*
- *Predictive distribution – 5 pts*

Total: 15 points

Solution

(a) **Posterior mean $\mathbf{m}_N$**

Design matrix and data:

$$\mathbf{\Phi} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}.$$

We have prior precision $\alpha = 2$ and noise precision $\beta = 10$.

Posterior formulas:

$$\mathbf{m}_N = \beta\, \mathbf{S}_N\, \mathbf{\Phi}^\top \mathbf{t} = 10\mathbf{S}_N \begin{pmatrix} 8 \\ 12 \\ 22 \end{pmatrix}$$

(b) **Posterior covariance $\mathbf{S}_N$**

$$\mathbf{S}_N = \left(\alpha\mathbf{I} + \beta\, \mathbf{\Phi}^\top \mathbf{\Phi}\right)^{-1}$$

Compute $\mathbf{\Phi}^\top\mathbf{\Phi}$:

$$\mathbf{\Phi}^\top\mathbf{\Phi} = \begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{pmatrix},$$

$$\beta\, \mathbf{\Phi}^\top\mathbf{\Phi} = 10 \cdot \begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{pmatrix} = \begin{pmatrix} 30 & 30 & 50 \\ 30 & 50 & 90 \\ 50 & 90 & 170 \end{pmatrix}.$$

$$\alpha\, \mathbf{I} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

$$\mathbf{S}_N^{-1} = \alpha\mathbf{I} + \beta\,\mathbf{\Phi}^\top\mathbf{\Phi} = \begin{pmatrix} 32 & 30 & 50 \\ 30 & 52 & 90 \\ 50 & 90 & 172 \end{pmatrix}.$$

$$\mathbf{S}_N = \frac{1}{12208}\begin{pmatrix} 844 & -660 & 100 \\ -660 & 3004 & -1380 \\ 100 & -1380 & 764 \end{pmatrix} \approx \begin{pmatrix} 0.0691 & -0.0541 & 0.0082 \\ -0.0541 & 0.2461 & -0.1130 \\ 0.0082 & -0.1130 & 0.0626 \end{pmatrix}.$$

Evaluating the Mean:

$$\mathbf{m}_N = 10\mathbf{S}_N\begin{pmatrix} 8 \\ 12 \\ 22 \end{pmatrix} = \frac{10}{12208}\begin{pmatrix} 1032 \\ 408 \\ 1048 \end{pmatrix} \approx \begin{pmatrix} 0.8453 \\ 0.3342 \\ 0.8585 \end{pmatrix}.$$

(c) **Predictive mean and variance at** $x_* = 1.5$

$$p(t_* \mid x_*, \mathbf{X}, \mathbf{t}) = \mathcal{N}\Big(\boldsymbol{\phi}(x_*)^\top\,\mathbf{m}_N,\ \frac{1}{\beta} + \boldsymbol{\phi}(x_*)^\top\,\mathbf{S}_N\,\boldsymbol{\phi}(x_*)\Big).$$

$$\boldsymbol{\phi}(x_*) = (1,\ 1.5,\ 2.25)^\top.$$

$$\bar{t}_* = \boldsymbol{\phi}(1.5)^\top\,\mathbf{m}_N \approx 1(0.8453) + 1.5(0.3342) + 2.25(0.8585) \approx 3.278.$$

$$\sigma_*^2 = 0.1 + \boldsymbol{\phi}(1.5)^\top\mathbf{S}_N\,\boldsymbol{\phi}(1.5) = 0.1 + \frac{625.75}{12208} \approx 0.1 + 0.0513 = 0.1513.$$

Model Evidence

Statement. We are comparing models using the Bayesian Marginal Likelihood (Model Evidence). We evaluate a model $\mathcal{M}$ with $M = 2$ parameters.

Assume a simplified 1D approximation for each parameter: before seeing data, the prior is completely flat with a width of $\Delta w_{\text{prior}} = 10$. After training, the posterior collapses into a tight uniform box of width $\Delta w_{\text{posterior}} = 0.1$. The likelihood of the training data evaluated at the MAP estimate is $p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) = 5.0$.

(a) Compute the log-likelihood of the data at the MAP estimate.

(b) Compute the complexity penalty term for this model.

(c) Sum the above together to approximate log marginal likelihood $\ln p(\mathcal{D}|\mathcal{M})$.

Grading:

- *Log-likelihood evaluation – 2 pts*
- *Complexity penalty evaluation – 5 pts*
- *Total log evidence – 3 pts*

Total: 10 points

Solution

(a) **Log-Likelihood:**

$$\ln p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) = \ln(5.0) \approx 1.609$$

(b) **Complexity Penalty (Occam Factor):** The penalty for M parameters is $M \ln\left(\frac{\Delta w_{\text{posterior}}}{\Delta w_{\text{prior}}}\right)$.

$$\text{Ratio} = \frac{0.1}{10} = 0.01$$

$$\text{Penalty} = 2\ln(0.01) \approx 2(-4.605) = -9.210$$

(c) **Approximate Log Marginal Likelihood:**

$$\ln p(\mathcal{D}|\mathcal{M}) \approx \ln p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) + M \ln\left(\frac{\Delta w_{\text{posterior}}}{\Delta w_{\text{prior}}}\right)$$

$$\ln p(\mathcal{D}|\mathcal{M}) \approx 1.609 - 9.210 = -7.601$$

Evidence Approximation: Hyperparameter Updates

Statement. We are training a Bayesian Linear Regression model with $M = 2$ features and performing the iterative Evidence Approximation to find the optimal hyperparameters.

At the current iteration, we have: Prior precision $\alpha = 1.0$, Noise precision $\beta = 2.0$. The static eigenvalues of the data design matrix product $\mathbf{\Phi}^T\mathbf{\Phi}$ are exactly $\lambda_1 = 4.0$ and $\lambda_2 = 1.0$. The current MAP weight vector is $\mathbf{m}_N = (0.5, 1.0)^T$.

(a) Calculate the complexity parameter γ (the effective number of well-determined parameters).

(b) Perform the M-step to calculate the updated hyperparameter α_{new}.

Grading:

- *Complexity parameter γ – 5 pts*
- *Updated α_{new} – 5 pts*

Total: 10 points

Solution

(a) **Complexity parameter γ:**

$$\gamma = \sum_{i=1}^{M} \frac{\beta\lambda_i}{\alpha + \beta\lambda_i} = \frac{8.0}{1.0 + 8.0} + \frac{2.0}{1.0 + 2.0}$$

$$\gamma = \frac{8}{9} + \frac{2}{3} = \frac{8}{9} + \frac{6}{9} = \frac{14}{9} \approx 1.555$$

(b) **Update α_{new}:** First, calculate the squared norm of the MAP weights:

$$\mathbf{m}_N^T\mathbf{m}_N = (0.5)^2 + (1.0)^2 = 0.25 + 1.0 = 1.25 = \frac{5}{4}$$

Now apply the update rule:

$$\alpha_{\text{new}} = \frac{\gamma}{\mathbf{m}_N^T\mathbf{m}_N} = \frac{14/9}{5/4} = \frac{14}{9} \times \frac{4}{5} = \frac{56}{45} \approx 1.244$$

Binary Classification via Least Squares Approximation

Statement. We have three data points in $\mathbb{R}^2$ with binary labels:

$$(x_1, t_1) = ((0,0), 0), \quad (x_2, t_2) = ((1,1), 1), \quad (x_3, t_3) = ((2,1), 1).$$

We attempt to fit them with a linear model

$$y(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2$$

by minimizing the sum of squared errors $\sum_{n=1}^{3} [t_n - y(x_n)]^2$. Then we classify by thresholding at $y(\mathbf{x}) = 0.5$.

(a) Write down the design matrix $\mathbf{\Phi}$.

(b) Solve the normal equations for $\mathbf{w}$.

(c) Discuss the implications of thresholding at 0.5 and compare with logistic regression.

Grading:

- *Design matrix – 4 pts*
- *Solve normal equations – 6 pts*
- *Threshold vs logistic regression – 5 pts*

Total: 15 points

Solution

(a) **Design matrix**

We form a row $(1,\ x_{n,1},\ x_{n,2})$ for each point:

$$\mathbf{\Phi} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

(b) **Solving the normal equations**

We compute $\mathbf{\Phi}^\top \mathbf{\Phi}$ and $\mathbf{\Phi}^\top \mathbf{t}$.

$$\mathbf{\Phi}^\top \mathbf{\Phi} = \begin{pmatrix} 3 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 2 \end{pmatrix}, \quad \mathbf{\Phi}^\top \mathbf{t} = \begin{pmatrix} (0+1+1) \\ (0+1\cdot 1+2\cdot 1) \\ (0+1\cdot 1+1\cdot 1) \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}.$$

Solving $(\mathbf{\Phi}^\top \mathbf{\Phi})\mathbf{w} = \mathbf{\Phi}^\top \mathbf{t}$ yields $\mathbf{w} = (0,\ 0,\ 1)$.

(c) **Thresholding at 0.5 & comparison with logistic regression**

The classification boundary is

$$w_0 + w_1 x_1 + w_2 x_2 = 0.5.$$

In practice, least squares does not directly account for the binary nature of t_n, so the resulting boundary may not separate classes well. Logistic regression, by contrast, models $p(t = 1 \mid \mathbf{x})$ and typically yields a more appropriate boundary for classification tasks.

Fisher's Linear Discriminant

Statement. We have a 2-class dataset in 2D space. The empirical means of the two classes are:

$$\mathbf{m}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

The total within-class covariance matrix has been computed as (its inverse is provided for convenience):

$$\mathbf{S}_W = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \qquad \mathbf{S}_W^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

(a) Determine Fisher's linear discriminant weight vector $\mathbf{w}$.

(b) Explain geometrically what the matrix $\mathbf{S}_W^{-1}$ is doing to the vector $(\mathbf{m}_2 - \mathbf{m}_1)$.

Grading:

- *Weight vector* $\mathbf{w}$ *– 5 pts*
- *Geometric explanation – 5 pts*

Total: 10 points

Solution

(a) **Fisher's Weight Vector $\mathbf{w}$:** Fisher's criterion yields a weight vector proportional to $\mathbf{S}_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1)$.

$$\mathbf{m}_2 - \mathbf{m}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\mathbf{w} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} (2)(2) + (-1)(2) \\ (-1)(2) + (2)(2) \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Since Fisher's discriminant is invariant to scaling, any vector proportional to this is a correct answer, such as $\mathbf{w} = (1, 1)^T$.

(b) **Geometric Explanation:** If the data clusters were perfectly spherical, projecting the data onto the line connecting the means $(\mathbf{m}_2 - \mathbf{m}_1)$ would be optimal. However, the non-diagonal elements of $\mathbf{S}_W$ indicate that the clusters are stretched ellipses. Multiplying by $\mathbf{S}_W^{-1}$ effectively rotates and scales the feature space to "un-stretch" the data, ensuring the projection plane perfectly separates the distributions rather than just pointing blindly at their centers.

Perceptron Learning Algorithm

Statement. We are training a Perceptron classifier on two data points. We append a bias term $x_0 = 1$ directly into our feature vectors $\boldsymbol{\phi}$. The target labels $t_n \in \{-1, +1\}$.

$$\boldsymbol{\phi}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad t_1 = +1$$

$$\boldsymbol{\phi}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad t_2 = -1$$

Our initial weight vector is $\mathbf{w}^{(0)} = (0, 0, 0)^T$. The learning rate is $\eta = 1$. The perceptron predicts $\hat{t} = \text{sign}(\mathbf{w}^T \boldsymbol{\phi})$. An update occurs if $t_n(\mathbf{w}^T \boldsymbol{\phi}_n) \leq 0$.

(a) Evaluate $\boldsymbol{\phi}_1$. Does it trigger an update? If so, calculate $\mathbf{w}^{(1)}$.

(b) Evaluate $\boldsymbol{\phi}_2$ using the new weights. Does it trigger an update? If so, calculate $\mathbf{w}^{(2)}$.

Grading:

- *Point 1 evaluation and update – 5 pts*
- *Point 2 evaluation and update – 5 pts*

Total: 10 points

Solution

(a) **Evaluate $\boldsymbol{\phi}_1$:** Compute the activation: $a_1 = (\mathbf{w}^{(0)})^T \boldsymbol{\phi}_1 = 0(1) + 0(0) + 0(1) = 0$. Check the condition: $t_1 a_1 = (1)(0) = 0$. Since $0 \leq 0$, it is misclassified. An update is triggered.

$$\mathbf{w}^{(1)} = \mathbf{w}^{(0)} + \eta t_1 \boldsymbol{\phi}_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + (1)(1) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

(b) **Evaluate $\boldsymbol{\phi}_2$:** Compute the activation: $a_2 = (\mathbf{w}^{(1)})^T \boldsymbol{\phi}_2 = 1(1) + 0(1) + 1(0) = 1$. Check the condition: $t_2 a_2 = (-1)(1) = -1$. Since $-1 \leq 0$, it is misclassified. An update is triggered.

$$\mathbf{w}^{(2)} = \mathbf{w}^{(1)} + \eta t_2 \boldsymbol{\phi}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + (1)(-1) \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

The new weight vector explicitly defines the decision boundary $0 - x_1 + x_2 = 0$.